Mathematics

Practice Workbook — Part 1

Contents

Sets

Sets

DAY 01

Set

A set is a well-defined collection of objects.

Examples:

- Odd natural numbers less than 10, i.e., 1, 3, 5, 7, 9
- The rivers of India
- The vowels in the English alphabet, namely, a, e, i, o, u

Symbols for the Special Sets

- $\mathbb{N}$: the set of all natural numbers
- $\mathbb{Z}$: the set of all integers
- $\mathbb{Q}$: the set of all rational numbers
- $\mathbb{R}$: the set of real numbers
- $\mathbb{Z}^+$: the set of positive integers
- $\mathbb{Q}^+$: the set of positive rational numbers
- $\mathbb{R}^+$: the set of positive real numbers.

Note

- Objects, elements and members of a set are synonymous terms.
- Sets are usually denoted by capital letters A, B, C, X, Y, Z, etc.
- The elements of a set are represented by small letters a, b, c, x, y, z, etc.
- If a is an element of a set A, we say that "a belongs to A"; the Greek symbol $\in$ (epsilon) is used to denote the phrase "belongs to". Thus, we write $a \in A$. If b is not an element of a set A, we write $b \notin A$ and read "b does not belong to A".

Classwork 1. Which of the following are sets? Justify your answer.
i) The collection of all the months of a year beginning with the letter J.
ii) The collection of ten most talented writers of India.
iii) A team of eleven best-cricket batsmen of the world. [NCERT]

Classwork 2. Let $A = \{1, 2, 3, 4, 5, 6\}$. Insert the appropriate symbol $\in$ or $\notin$ in the blank spaces:
i) $5 \in A$ ii) $8 \notin A$ iii) $0 \notin A$ [NCERT]

Note

◇ In roster form, every element of the set is listed only once.
◇ The order in which the elements are listed is immaterial. For example, each of the following sets denotes the same set $\{1, 2, 3\}$, $\{3, 2, 1\}$, $\{1, 3, 2\}$.

Representation of a Set

Usually, sets are represented in the following two ways:

1. Roster form or Tabular form
2. Set-Builder form or Rule Method

Roster form (Tabular form)

In this form, we list all the members of the set within braces (curly brackets) and separate these by commas.

Example

The set A of all odd natural numbers less than 10 in the Roster form is written as:

$$A = \{1, 3, 5, 7, 9\}$$

Classwork 3. $A = \{x : x \text{ is a natural number } < 3\}$. Write A in roster form.

Classwork 4. Consider the set $A = \{x : x \text{ is an integer}, 0 \leq x < 4\}$. Write A in Roster form.

Classwork 5. Let $A = \{x : x \text{ is a prime number } < 6\}$. Write A in roster form.

Classwork 6. Write the following set in roster form. $A =$ The set of all letters in the word TRIGONOMETRY.

Set-Builder Form

In this form, we write a variable (say x) representing any member of the set followed by a property satisfied by each member of the set.

Example

The set A of all prime numbers less than 10 in the set-builder form is written as

$$A = \{x \mid x \text{ is a prime number less than } 10\}.$$

Note

The symbol '|' stands for the words "such that". Sometimes, we use the symbol ':' in place of the symbol '|'.

Classwork 7. Write the set $\left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \frac{6}{7}\right\}$ in the set-builder form. [NCERT]

Classwork 8. Write the following set in the set-builder form:
i) $\{3, 6, 9, 12\}$ ii) $\{2, 4, 8, 16, 32\}$ [NCERT]

Homework 1. Write the set $A = \{1, 4, 9, 16, 25, \ldots\}$ in set-builder form. [NCERT]

Classwork 9. Match each of the set on the left described in the roster form with the same set on the right described in the set-builder form:

(i)	{P, R, I, N, C, A, L}	(a)	$\{x : x \text{ is a positive integer and is a divisor of } 18\}$
(ii)	$\{0\}$	(b)	$\{x : x \text{ is an integer and } x^2 - 9 = 0\}$
(iii)	$\{1, 2, 3, 6, 9, 18\}$	(c)	$\{x : x \text{ is an integer and } x + 1 = 1\}$
(iv)	$\{3, -3\}$	(d)	$\{x : x$ is a letter of the word PRINCIPAL$\}$

[NCERT]

Homework 2. List all the elements of the following sets:
i) $A = \{x : x \text{ is an odd natural number}\}$

ii) $B = \{x : x \text{ is an integer, } -\frac{1}{2} < x < \frac{9}{2}\}$
iii) $C = \{x : x \text{ is a letter in the word "LOYAL"}\}$ [NCERT]

Sets

DAY 02

Types of Sets

Empty set or Null set

A set which has no element is called the null set or empty set. It is denoted by the symbol ϕ.

Examples

- The set of all real numbers whose square is -1.
- The set of all rational numbers whose square is 2.
- The set of all those integers that are both even and odd.

Note

A set consisting of at least one element is called a non-empty set.

Is $\{\phi\}$ an empty set?

Classwork 10. Which of the following are examples of the null set?
i) Set of odd natural numbers divisible by 2
ii) Set of even prime numbers [NCERT]

Cardinality of a Set

The number of elements in a finite set is represented by $n(A)$, known as the *cardinality* of the set.

Classwork 11. If $A = \{x : x \text{ is a natural number, } x < 5 \text{ and } x > 7\}$, then $n(A)$ is
(A) 1 (B) 0 (C) 2 (D) 3 [NCERT]

Singleton set

A set having only one element is called a singleton set.

Example

$\{0\}$ is a singleton set, whose only member is 0.

Finite and Infinite set

A set which has a finite number of elements is called a finite set. Otherwise, it is called an infinite set.

Example

The set of all days in a week is a finite set whereas the set of all integers, denoted by $\{\ldots, -2, -1, 0, 1, 2, \ldots\}$ or $\{x \mid x \text{ is an integer}\}$, is an infinite set.

Note

All infinite sets cannot be described in the roster form. For example, the set of real numbers cannot be described in this form, because the elements of this set do not follow any particular pattern.

Classwork 12. Which of the following sets are finite or infinite?
i) The set of months of a year
ii) $\{1, 2, 3, \ldots\}$
iii) $\{1, 2, 3, \ldots 99, 100\}$
iv) The set of prime numbers less than 99 [NCERT]

Homework 3. In the following, state whether $A = B$ or not:
(i) $A = \{a, b, c, d\}$, $B = \{d, c, b, a\}$
(ii) $A = \{4, 8, 12, 16\}$, $B = \{8, 4, 16, 18\}$
(iii) $A = \{2, 4, 6, 8, 10\}$, $B = \{x : x \text{ is a positive even integer and } x \leq 10\}$
(iv) $A = \{x : x \text{ is a multiple of } 10\}$, $B = \{10, 15, 20, 25, 30, \ldots\}$

Equal sets

Two sets A and B are said to be equal, written as $A = B$, if every element of A is in B and every element of B is in A.

Note

A set does not change if one or more elements of the set are repeated. For example, the sets $A = \{1, 2, 3\}$ and $B = \{2, 2, 1, 3, 3\}$ are equal, since each element of A is in B and vice-versa. That is why we generally do not repeat any element in describing a set.

Classwork 13. Find the pairs of equal sets, if any; give reasons:
$A = \{0\}, \quad B = \{x : x - 5 = 0\}, \quad D = \{x : x^2 = 25\}.$ [NCERT]

Equivalent sets

Two finite sets A and B are said to be equivalent, if $n(A) = n(B)$. Clearly, equal sets are equivalent but equivalent sets need not be equal.

Example

The sets $A = \{4, 5, 3, 2\}$ and $B = \{1, 6, 8, 9\}$ are equivalent but are not equal.

Subset

Let A and B be two sets. If every element of A is an element of B, then A is called a subset of B and we write $A \subset B$ or $B \supset A$ (read as "A is contained in B" or "B contains A"). B is called the superset of A.

Note

- Every set is a subset and a superset of itself.
- If A is not a subset of B, we write $A \not\subset B$.
- The empty set is the subset of every set.
- If A is a set with $n(A) = m$, then the number of subsets of A is 2^m and the number of proper subsets of A is $2^m - 1$.

Example

Let $A = \{3, 4\}$, then the subsets of A are ϕ, $\{3\}$, $\{4\}$, $\{3, 4\}$.
Here, $n(A) = 2$ and number of subsets of $A = 2^2 = 4$.

Also, $\{3\} \subset \{3,4\}$ and $\{2,3\} \not\subset \{3,4\}$.

Classwork 14. Let $A = \{1,2,3\}$. Then the no. of subsets of A is
(A) 3 (B) 6 (C) 8 (D) 9

Classwork 15. Consider $A = \{x : x \text{ is an integer}, 0 < x < 4\}$.
i) Write A in roster form.
ii) The number of proper subsets of A is ...

Classwork 16. If the number of proper subsets of a set is 63, then the number of elements of the set is ......

Classwork 17. Write all subsets of $\{1,2,3\}$. [NCERT]

Classwork 18. $A = \{x : x \text{ is a natural number less than } 8\}$
i) Write A in roster form.
ii) Write a subset of A containing all even numbers in A.

Classwork 19. Consider the sets $A = \{1,3\}$, $B = \{1,5,9\}$, $C = \{1,3,5,7,9\}$ and ϕ. Insert the symbol $\subset$ or $\not\subset$ between each of the following pair of sets:
i) $\phi \subset B$ ii) $A \not\subset B$ iii) $A \subset C$ iv) $B \subset C$ [NCERT]

Sets

DAY 03

Homework 4. Examine whether the following statements are true or false:
i) $\{a, b\} \subset \{b, c, a\}$
ii) $\{a, e\} \subset \{x : x \text{ is a vowel in the English alphabet}\}$
iii) $\{1, 2, 3\} \subset \{1, 3, 5\}$
iv) $\{a\} \subset \{a, b, c\}$
v) $\{a\} \in \{a, b, c\}$
vi) $\{x : x \text{ is an even natural number less than } 6\} \subset \{x : x \text{ is a natural number which divides } 36\}$

Note

- $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}$
- $\mathbb{Q} \subset \mathbb{R}$
- $\mathbb{T} \subset \mathbb{R}$
- $\mathbb{N} \not\subset \mathbb{T}$

Intervals as Subset of $\mathbb{R}$

The interval which contains the end points also is called a closed interval and is denoted by $[a, b]$. Thus

$$[a, b] = \{x : a \leq x \leq b\}.$$

We can also have intervals closed at one end and open at the other, i.e.,

$[a, b) = \{x : a \leq x < b\}$ is an interval from a to b, including a but excluding b.
$(a, b] = \{x : a < x \leq b\}$ is an interval from a to b including b but excluding a.
$(a, b) = \{x : a < x < b\}$ is an interval from a to b excluding both a and b.

Classwork 20. Which one of the following is equal to $\{x : x \in \mathbb{R},\ -4 < x \leq 5\}$?
(A) $(-4, 5]$ (B) $(-4, 5)$ (C) $[-4, -5]$ (D) $[-4, 5)$

Classwork 21. The set-builder form of $(6, 12)$ is
(A) $\{x : x \in \mathbb{R},\ 6 < x \leq 12\}$
(B) $\{x : x \in \mathbb{R},\ 6 < x < 12\}$
(C) $\{x : x \in \mathbb{R},\ 6 \leq x \leq 12\}$
(D) $\{x : x \in \mathbb{R},\ 6 \leq x < 12\}$

Universal Set

A universal set (usually denoted by U) is a set which has elements of all the related sets, without any repetition of elements.

Example

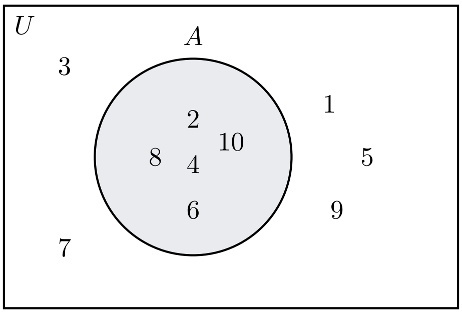
If A and B are two sets, such as $A = \{1, 2, 3\}$ and $B = \{1, a, b, c\}$, then the universal set associated with these two sets is given by $U = \{1, 2, 3, a, b, c\}$.

Classwork 22. Given the sets $A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$ and $C = \{0, 2, 4, 6, 8\}$, which of the following may be considered as universal set(s) for all the three sets A, B and C?
(A) $\{0, 1, 2, 3, 4, 5, 6\}$
(B) ϕ

(C) $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
(D) $\{1, 2, 3, 4, 5, 6, 7, 8\}$ [NCERT]

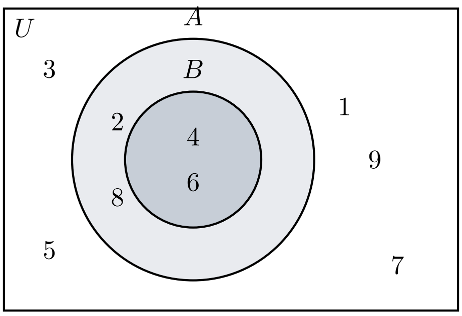
Venn Diagram

Most of the relationships between sets can be represented by means of diagrams which are known as Venn diagrams. The universal set is represented usually by a rectangle and its subsets by circles.



Note

- In the figure, $U = \{1, 2, 3, \ldots, 10\}$ is the universal set of which $A = \{2, 4, 6, 8, 10\}$ is a subset.
- $U = \{1, 2, 3, \ldots, 10\}$ is the universal set of which $A = \{2, 4, 6, 8, 10\}$ and $B = \{4, 6\}$ are subsets, and also $B \subset A$.

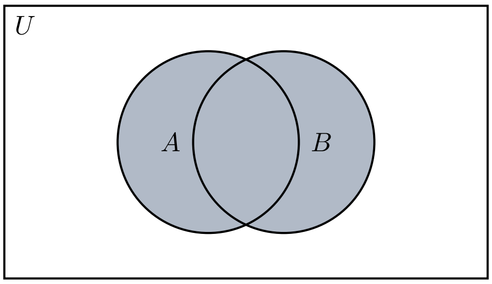


Operations on Sets

Union of sets

The union of two sets A and B is the set C which consists of all those elements which are either in A or in B (including those which are in both). In symbols, we write,

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$



Some Properties of the Operation of Union

- $A \cup B = B \cup A$ (Commutative law)
- $(A \cup B) \cup C = A \cup (B \cup C)$ (Associative law)
- $A \cup \phi = A$ (Law of identity element, ϕ is the identity of U)
- $A \cup A = A$ (Idempotent law)
- $U \cup A = U$ (Law of U)

Classwork 23. Find the union of each of the following pairs of sets:
(i) $X = \{1, 3, 5\}$, $Y = \{1, 2, 3\}$
(ii) $A = \{a, e, i, o, u\}$, $B = \{a, b, c\}$ [NCERT]

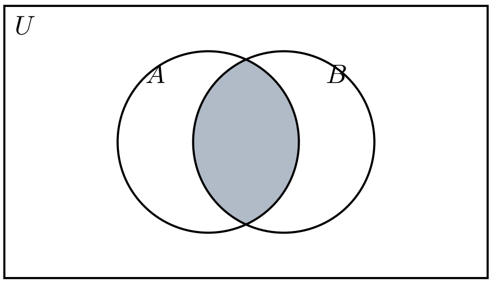
Classwork 24. If A and B are two sets such that $A \subset B$, then $A \cup B$ is
(A) A (B) Null set (C) B (D) $\{\phi\}$

Homework 5. If $A = \{x : x \text{ is a factor of } 6\}$, $B = \{x : x \text{ is an even integer, } 2 \leq x < 9\}$, then find $A \cup B$.

Intersection of sets

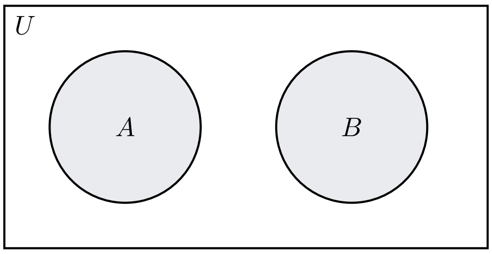
The intersection of two sets A and B is the set of all those elements which belong to both A and B. Symbolically, we write

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$



Disjoint sets

If A and B are two sets such that $A \cap B = \phi$, then A and B are called disjoint sets.



Example

Let $A = \{2, 4, 6, 8\}$ and $B = \{1, 3, 5, 7\}$. Then A and B are disjoint sets, because there are no elements which are common to A and B.

Some Properties of Operation of Intersection

- $A \cap B = B \cap A$ (Commutative law)

- $(A \cap B) \cap C = A \cap (B \cap C)$ (Associative law)
- $\phi \cap A = \phi$, $U \cap A = A$ (Law of ϕ and U)
- $A \cap A = A$ (Idempotent law)
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (Distributive law, i.e., $\cap$ distributes over $\cup$)

Classwork 25. Let $A = \{x : x \text{ is a natural number less than } 6\}$ and $B = \{1, 2\}$.
i) Write A in roster form.
ii) Find $A \cap B$.

Classwork 26. If $A = \{2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7\}$, then write
i) $A \cup B$ ii) $A \cap B$

Note

If U is the universal set and A is any set, then $U \cup A = U$.

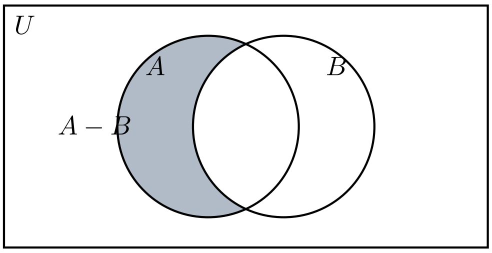
Homework 6. If U is the universal set and A is any set, then $U \cap A =$
(A) U (B) A (C) ϕ (D) None of these.

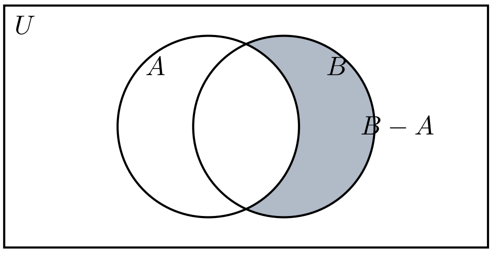
Classwork 27. Which of the following pairs of sets are disjoint?
(i) $\{1, 2, 3, 4\}$ and $\{x : x \text{ is a natural number and } 4 \le x \le 6\}$
(ii) $\{a, e, i, o, u\}$ and $\{c, d, e, f\}$
(iii) $\{x : x \text{ is an even integer}\}$ and $\{x : x \text{ is an odd integer}\}$ [NCERT]

Classwork 28. If two sets A and B are disjoint, which one among the following is true?
(A) $A \cup B = A$ (B) $A \cup B = B$
(C) $A \cap B = B$ (D) $A \cap B = \phi$

Difference of Sets

The difference of the sets A and B in this order is the set of elements which belong to A but not to B. Symbolically, we write $A - B$ and read as "A minus B".





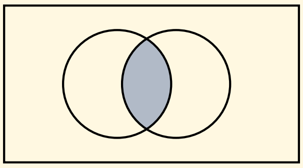
Classwork 29. Let $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 8\}$. Find $A - B$ and $B - A$.

Classwork 30. If $X = \{a, b, c, d\}$ and $Y = \{f, b, d, g\}$, find
(i) $X - Y$ (ii) $Y - X$ (iii) $X \cap Y$ [NCERT]

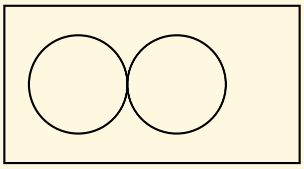
Classwork 31. If $A = \{x : x \text{ is a letter in the word "MATHEMATICS"}\}$ and $B = \{y : y \text{ is a letter in the word "STATISTICS"}\}$, then find $A - B$. [NCERT]

Note

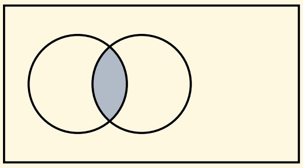
Distributive law visualised: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$



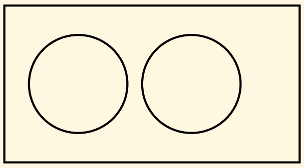
$B \cap C$



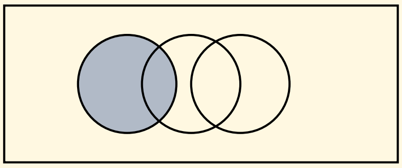
$A \cap (B \cap C)$



$A \cap B$



$A \cap C$



$(A \cap B) \cup (A \cap C)$

Sets

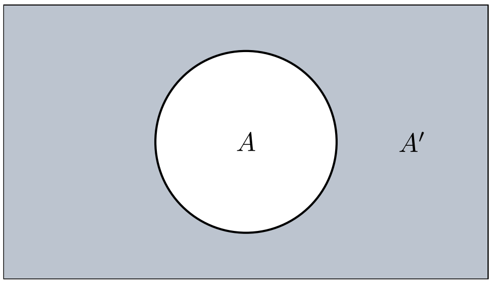
DAY 04

Complement of a Set

Let U be the universal set and A a subset of U. Then the complement of A is the set of all elements of U which are not the elements of A. Symbolically, we write A' to denote the complement of A with respect to U. Thus,

$$A' = \{x : x \in U \text{ and } x \notin A\}.$$

Obviously $A' = U - A$.



Classwork 32. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $A = \{1, 3, 5, 7, 9\}$. Find A'. [NCERT]

Classwork 33. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$ and $C = \{3, 4, 5, 6\}$. Find
(i) A' (ii) B' (iii) $(A \cup C)'$ (iv) $(A \cup B)'$ (v) $(A')'$ [NCERT]

Homework 7. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{2, 4, 6, 8\}$, $B = \{2, 4, 8\}$.
i) Find A' and B'.
ii) Find $(A \cup B)'$.

Classwork 34. Draw Venn diagrams which represent
i) $(A \cup B)'$ ii) $A' \cup B'$

Classwork 35. If P and Q are two sets such that $n(P) = 12$, $n(Q - P) = 7$, then which among the following is the value of $n(P \cup Q)$?
(A) 12 (B) 7 (C) 19 (D) 5

Classwork 36. If $M \subset N$, then draw the Venn diagram of $N - M$.

De Morgan's Law

The complement of the union of two sets is the intersection of their complements, and the complement of the intersection of two sets is the union of their complements.

De Morgan's laws:

$$(A \cup B)' = A' \cap B'$$
$$(A \cap B)' = A' \cup B'$$

Classwork 37. Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{2, 3\}$ and $B = \{3, 4, 5\}$. Find A', B', $A' \cap B'$, $A \cup B$ and hence show that $(A \cup B)' = A' \cap B'$. [NCERT]

Classwork 38. $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$
$A = \{x : x \in U \text{ and } x \text{ is a prime number}\}$

$B = \{x : x \in U \text{ and } x > 2\}$
i) Write A and B in roster form.
ii) Verify that $(A \cap B)' = A' \cup B'$.

Classwork 39. The set $(A \cap B')' \cup (B \cap C)$ is equal to
(A) $A' \cup B \cup C$ (B) $A' \cup B$ (C) $A' \cup C'$ (D) $A' \cap B$ (E) $A' \cup C$

Homework 8. If X and Y are sets and X' denotes the complement of X, then $X \cap (X \cup Y)'$ is equal to
(A) X (B) Y (C) ϕ (D) $X \cap Y$ (E) $X \cup Y$

Some Properties of Complement of Sets

Complement laws

(i) $A \cup A' = U$

(ii) $A \cap A' = \phi$

Law of Double Complementation

$$(A')' = A$$

Laws of Empty set and Universal set

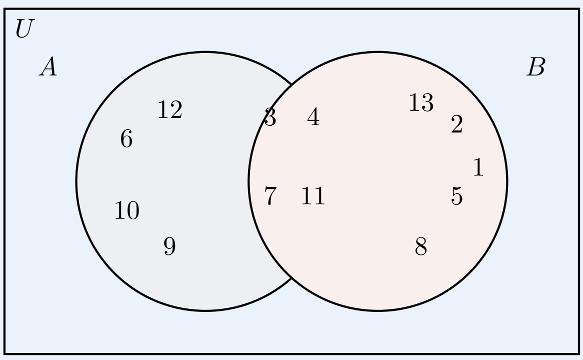
$$\phi' = U \quad \text{and} \quad U' = \phi.$$

These laws can be verified by using Venn diagrams.

Classwork 40. Fill in the blanks to make each of the following a true statement:
(i) $A \cup A' = \ldots$ (ii) $\phi' \cap A = \ldots$
(iii) $A \cap A' = \ldots$ (iv) $U' \cap A = \ldots$ [NCERT]

Homework 9. If A is any set, then $A \cap A' =$
(A) A (B) ϕ (C) A' (D) U

Classwork 41. Consider a Venn diagram of $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\}$
i) Write sets A, B in Roster form.
ii) Verify $(A \cup B)' = A' \cap B'$.
iii) Find $n(A \cap B)'$.



Homework 10. Let $U = \{x : x \text{ is an integer}, \ -4 < x < 4\}$ be the universal set. $A = \{x : x \text{ is an integer}, 0 \leq x \leq 3\}$ and $B = \{x : x \text{ is an integer}, \ -3 < x < 1\}$ are the subsets of U.
a) Write A and B in the roster form.
b) Verify $(A \cup B)' = A' \cap B'$.

Try Yourself

1. Write the set $\{x : x \text{ is a positive integer and } x^2 < 40\}$ in the Roster form.
2. Match each of the set on the left in the roster form with the same set on the right described in set-builder form:

(i)	$\{1, 2, 3, 6\}$	(a)	$\{x : x$ is a prime number and a divisor of $6\}$
(ii)	$\{2, 3\}$	(b)	$\{x : x$ is an odd natural number less than $10\}$
(iii)	$\{M, A, T, H, E, I, C, S\}$	(c)	$\{x : x$ is a natural number and divisor of $6\}$
(iv)	$\{1, 3, 5, 7, 9\}$	(d)	$\{x : x$ is a letter of the word MATHEMATICS$\}$

3. (a) Write the set-builder form of the subset $(-1, 2)$ of $\mathbb{R}$.
(b) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ be the universal set having subsets $A = \{1, 3, 5, 7, 9\}$ and $B = \{5, 6, 7, 8, 9, 10\}$, then find
(i) $A \cup B$ (ii) A' and B' (iii) Verify that $(A \cup B)' = A' \cap B'$.
4. Let $A = \{x : x \in \mathbb{W}, \ x < 5\}$ and $B = \{x : x \text{ is a prime number less than } 5\}$, $U = \{x : x \text{ is an integer}, \ 0 \leq x \leq 6\}$.
i) Write A, B in roster form.
ii) Find $(A - B) \cup (B - A)$.
iii) Verify that $(A \cup B)' = A' \cap B'$.
5. i) Number of proper subsets of a set with 4 elements is ...
ii) Consider the sets $A = \{x : x$ is a letter in the word MATHEMATICS$\}$ and $B = \{y : y$ is a letter in the word PHYSICS$\}$. Then find
(a) $A \cup B$ (b) $A \cap B$ (c) $A - B$.

6. If $A = \{x : x \in \mathbb{R}, \ -2 \leq x \leq 5\}$ and $B = \{x : x \in \mathbb{R}, \ 2 < x \leq 6\}$, then which among the following is $A \cap B$?
(A) $[2, 5]$ (B) $(2, 5]$ (C) $[2, 5)$ (D) $[-2, 6]$

8. If $A = \{x : x \text{ is a natural number}\}$, $B = \{x : x \text{ is an even natural number}\}$, $C = \{x : x \text{ is an odd natural number}\}$ and $D = \{x : x \text{ is a prime number}\}$, find
(i) $A \cap B$ (ii) $A \cap C$ (iii) $A \cap D$
(iv) $B \cap C$ (v) $B \cap D$ (vi) $C \cap D$ [NCERT]

9. i) For two sets A and B, $A \cup B = A$ iff
(A) $A = \phi$ (B) $A \neq B$ (C) $A \subseteq B$ (D) $B \subseteq A$
ii) Write all subsets of $\{-1, 1\}$.
iii) Write the set-builder form of $\{3, 6, 9, 12, 15\}$. [NCERT]

7. Let $A = \{x : x \text{ is an integer}, \ \frac{1}{2} < x < \frac{7}{2}\}$, $B = \{2, 3, 4\}$.
a) Write A in roster form.
b) Verify that $(A - B) \cup (A \cap B) = A$.

10. Write the following as intervals:
i) $\{x : x \in \mathbb{R}, \ -4 < x \leq 6\}$
ii) $\{x : x \in \mathbb{R}, \ -12 < x < -10\}$

iii) $\{x : x \in \mathbb{R},\ 0 \leq x < 7\}$
iv) $\{x : x \in \mathbb{R},\ 3 \leq x \leq 4\}$ [NCERT]

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Relations and Functions

Relations and Functions

DAY 01

Ordered Pair

An ordered pair is a pair of entries in the specified order. The two entries are separated by a comma and enclosed with brackets. In (a, b), a is the first element and b is the second element.

Cartesian Product of Sets

Given two non-empty sets P and Q. The cartesian product $P \times Q$ is the set of all ordered pairs of elements from P and Q, i.e.,

$$P \times Q = \{(p, q) : p \in P,\ q \in Q\}$$

Classwork 1. Let $A = \{1, 2, 3\}$ and $B = \{3, 4\}$. Find $A \times B$.

Classwork 2. If $A = \{a, b, c\}$, $B = \{x, y\}$, then write $A \times B$.

Classwork 3. If $A \times B = \{(p, q), (p, r), (m, q), (m, r)\}$, find A and B. [NCERT]

Note

◇ If either P or Q is the null set, then $P \times Q$ will also be the empty set, i.e., $P \times Q = \phi$.

◇ If A and B are non-empty sets and either A or B is an infinite set, then so is $A \times B$.

Think

Is $A \times B = B \times A$?

Classwork 4. If $G = \{7, 8\}$ and $H = \{5, 2, 4\}$. Find $G \times H$ and $H \times G$. [NCERT]

Homework 1. If $P = \{a, b, c\}$ and $Q = \{r\}$, form the sets $P \times Q$ and $Q \times P$. Are these two products equal? [NCERT]

Note

If there are p elements in A and q elements in B, then there will be pq elements in $A \times B$, i.e., if $n(A) = p$ and $n(B) = q$, then $n(A \times B) = pq$.

Classwork 5. If the set A has 3 elements and the set $B = \{3, 4, 5\}$, then find the number of elements in $(A \times B)$. [NCERT]

$n(A \times B) = n(A) \cdot n(B)$

$= 3 \times 3 = 9$

Classwork 6. Let $A = \{1, 2\}$ and $B = \{3, 4\}$. Write $A \times B$. How many subsets will $A \times B$ have? [NCERT]

Classwork 7. If $A = \{3, 4\}$, form the set $A \times A$.

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Relations and Functions

DAY 02

Note

$A \times A \times A = \{(a, b, c) : a, b, c \in A\}$. Here (a, b, c) is called an ordered triplet.

Classwork 8. If $A = \{-1, 1\}$, find $A \times A \times A$. [NCERT]

Homework 2. If $P = \{1, 2\}$, form the set $P \times P \times P$. [NCERT]

Classwork 9. Let $A = \{1, 2, 3\}$, $B = \{3, 4\}$ and $C = \{4, 5, 6\}$. Find
(i) $A \times (B \cap C)$ (ii) $(A \times B) \cap (A \times C)$
(iii) $A \times (B \cup C)$ (iv) $(A \times B) \cup (A \times C)$

Classwork 10. State whether each of the following statements are true or false. If the statement is false, rewrite the given statement correctly.
(i) If A and B are non-empty sets, then $A \times B$ is a non-empty set of ordered pairs (x, y) such that $x \in A$ and $y \in B$.
(ii) If $A = \{1, 2\}$, $B = \{3, 4\}$, then $A \times (B \cap \phi) = \phi$. [NCERT]

Note

Two ordered pairs are equal if and only if the corresponding first elements are equal and the second elements are also equal.

Classwork 11. If $(x + 1,\ y - 2) = (3, 1)$, then find the values of x and y. [NCERT]

Classwork 12. If $\left(\frac{x}{3} + 1,\ y - \frac{2}{3}\right) = \left(\frac{5}{3}, \frac{1}{3}\right)$, find the values of x and y. [NCERT]

Homework 3. If $(x + 1,\ y - 4) = (3, 7)$, then find the values of x and y.

Classwork 13. The Cartesian product $A \times A$ has 9 elements among which 2 elements are $(-a, 0)$ and $(0, a)$. Write A. Also find $A \times A$.

Homework 4. The Cartesian product $A \times A$ has 9 elements among which are found $(-1, 0)$ and $(0, 1)$. Find the set A and the remaining elements of $A \times A$. [NCERT]

Relations and Functions

DAY 03

Relations

A relation R from a non-empty set A to a non-empty set B is a subset of the cartesian product $A \times B$. The subset is derived by describing a relationship between the first element and the second element of the ordered pairs in $A \times B$. The second element is called the *image* of the first element.

Note

- A relation may be represented algebraically either by the Roster method or by the Set-builder method.
- An arrow diagram is a visual representation of a relation.

Note

- Relation $R \subseteq A \times B$ means $R : A \to B$
- Relation defined from A to B means $R : A \to B$
- Relation defined on A means $R : A \to A$

Classwork 14. $A = \{1, 2, 3, 5\}$ and $B = \{4, 6, 9\}$. Define a relation R from A to B by $R = \{(x, y) :$ the difference between x and y is odd; $x \in A$, $y \in B\}$. Write R in roster form.

Classwork 15. Write the relation $R = \{(x, x^3) : x$ is a prime number $< 10\}$, in roster form. [NCERT]

Domain, Range and Codomain

Domain. The set of all first elements of the ordered pairs in a relation R from a set A to a set B is called the *domain* of the relation R.

Range

The set of all second elements in a relation R from a set A to a set B is called the *range* of the relation R.

Codomain

The whole set B is called the *codomain* of the relation R. Note that range $\subseteq$ codomain.

Example

$A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$.
R be a relation defined from A to B such that $R = \{(x, y) : x + y = 8\}$
$\Rightarrow R = \{(2, 6), (3, 5)\}$
Then, Domain $= \{2, 3\}$, Range $= \{5, 6\}$, Codomain $= B = \{4, 5, 6\}$.

Classwork 16. R is a relation defined on the set $A = \{1, 2, 3, \ldots, 14\}$ by $R = \{(x, y) : 3x - y = 0;\ x, y \in A\}$. Write the domain, co-domain and the range. [NCERT]

Classwork 17. A relation R on the set of natural numbers is defined by $R = \{(x, y) : y = x;\ x$ is a natural number less than 4, $x, y \in \mathbb{N}\}$
(i) Write the relation in Roster form.
(ii) Write the domain and range of the relation.

Classwork 18. Let $A = \{1, 2, 3, 4, 6\}$ and R be a relation on A defined by $R = \{(a, b) : a, b \in A;\ b$ is exactly divisible by $a\}$
(i) Write R in the roster form.

(ii) Find the domain and range of R. [NCERT]

Classwork 19. Let R be the relation on $\mathbb{Z}$ defined by $R = \{(a, b) : a, b \in \mathbb{Z},\ a - b \text{ is an integer}\}$. Find the domain and range of R. [NCERT]

Classwork 20. Define a relation R on the set $\mathbb{N}$ of natural numbers by $R = \{(x, y) : y = x + 5,\ x \text{ is a natural number less than } 4;\ x, y \in \mathbb{N}\}$. Depict this relationship using roster form. Write down the domain and range. [NCERT]

Homework 5. Determine the domain and range of the relation R defined by $R = \{(x, x + 5) : x \in \{0, 1, 2, 3, 4, 5\}\}$. [NCERT]

Relations and Functions

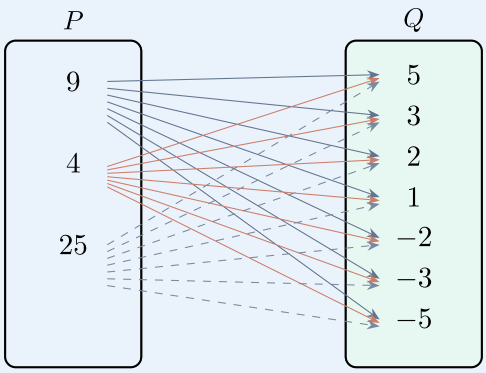
DAY 04

Arrow Diagram

$R = \{(x, y) : x$ is the first letter of the name $y,\ x \in P,\ y \in Q\}$.
$R = \{(a, \text{Ali}),\ (b, \text{Bhanu}),\ (b, \text{Binoy}),\ (c, \text{Chandra})\}$

Classwork 21. Let $A = \{1, 2, 3, 4, 5, 6\}$. Define a relation R from A to A by $R = \{(x, y) : y = x + 1\}$
(i) Depict this relation using an arrow diagram.
(ii) Write down the domain, codomain and range of R. [NCERT]

Classwork 22. The following figure shows a relation between the set P and Q. Write this relation
(i) in set-builder form
(ii) in roster form
What is its domain and range?



[NCERT]

Note

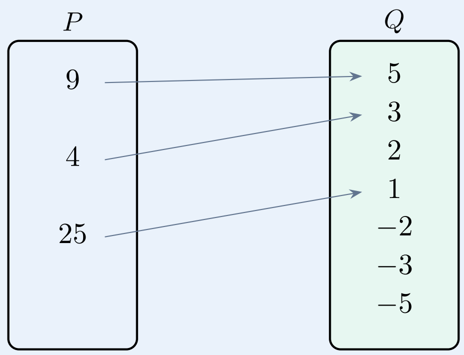
The total number of relations that can be defined from a set A to a set B is the number of possible subsets of $A \times B$. If $n(A) = p$ and $n(B) = q$, then $n(A \times B) = pq$ and the total number of relations is 2^{pq}.

Classwork 23. Let $A = \{x, y, z\}$ and $B = \{1, 2\}$. Find the number of relations from A to B. [NCERT]

Classwork 24. Let $A = \{1, 2\}$ and $B = \{3, 4\}$. Find the number of relations from A to B. [NCERT]

Homework 6. $A = \{2, 3\}$, $B = \{1, 3, 5\}$, then the number of relations from A to B is
(A) 2 (B) 32 (C) 64 (D) 62

Classwork 25. Determine whether the following figure is a function or not.



Functions

A relation f from a set A to a set B is said to be a function if every element of set A has one and only one image in set B. The function f from A to B is denoted by $f : A \to B$.

Classwork 26. Examine each of the following relations given below and state in each case, giving reasons, whether it is a function or not.
(i) $R = \{(2,1),(3,1),(4,2)\}$
(ii) $R = \{(2,2),(2,4),(3,3),(4,4)\}$
(iii) $R = \{(1,2),(2,3),(3,4),(4,5),(5,6),(6,7)\}$ [NCERT]

Domain, Codomain, Range

The set A is called the *domain* of f and the set B is called the *codomain* of f. The set of elements of B which are the images of the elements of A are called the *range* of f.

Classwork 27. Let $A = \{1,2,3,4\}$, $B = \{1,5,9,11,15,16\}$ and $f = \{(1,5),(2,9),(3,1),(4,5),(2,11)\}$. State with the reason, whether f is a relation or a function. [NCERT]

Image and Pre-Image

Let $f : A \to B$ be a function and $(x,y) \in f$, then y is the *image* of x under the function f, denoted by $y = f(x)$, and x is called the *pre-image* of y.

Classwork 28. Let N be the set of natural numbers and the relation R be defined on N such that $R = \{(x,y) : y = 2x,\ x,y \in \mathbb{N}\}$. What is the domain, codomain and range of R? Is this relation a function? [NCERT]

Homework 7. Let R be a relation defined by $R = \{(1,3),(1,5),(2,5)\}$. State whether R is a function or not. Justify your answer. [NCERT]

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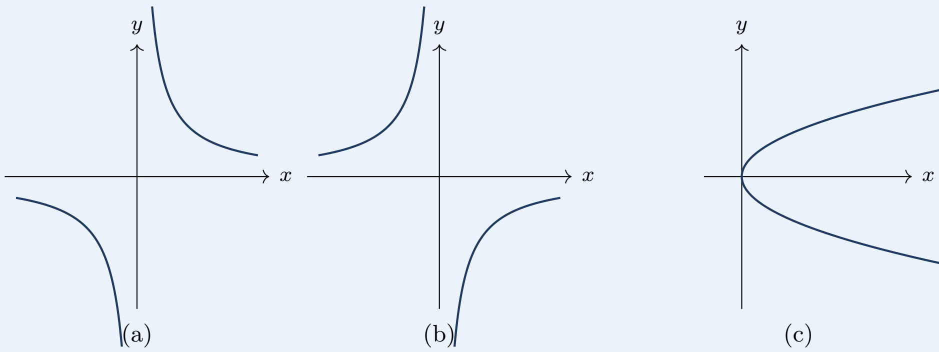
Relations and Functions

DAY 05

Vertical Line Test

The vertical line test can be used to determine whether a graph represents a function. A vertical line includes all points with a particular x-value. If we can draw any vertical line that intersects a graph more than once, then the graph does not define a function because that x value has more than one y-value.

Classwork 29. Consider the following graphs:



(a) (b) (c)

Which graph does not represent a function?

Value of a Function at a Point

Classwork 30. A function f is defined by $f(x) = 2x - 5$. Write down the values of
(i) $f(0)$ (ii) $f(7)$ (iii) $f(-3)$ [NCERT]

Homework 8. Let N be the set of natural numbers. Define a real valued function $f : N \to N$ by $f(x) = 2x + 1$. Using this definition, complete the table given below.

x	y
1	$f(1) = \ldots$
2	$f(2) = \ldots$
3	$f(3) = \ldots$
4	$f(4) = \ldots$
5	$f(5) = \ldots$
6	$f(6) = \ldots$
7	$f(7) = \ldots$

[NCERT]

Real Valued Function and Real Constant Function

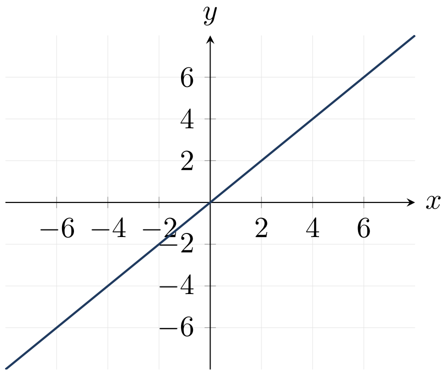
A function which has either $\mathbb{R}$ or one of its subsets as its range is called a *real valued function*. Further, if its domain is also either $\mathbb{R}$ or a subset of $\mathbb{R}$, it is called a *real function*.

Classwork 31. Let $f = \{(1, 1), (2, 3), (0, -1), (-1, -3)\}$ be a function from $\mathbb{Z}$ to $\mathbb{Z}$ defined by $f(x) = ax + b$, for some integers a and b. Determine a and b. [NCERT]

Some Functions and their Graphs

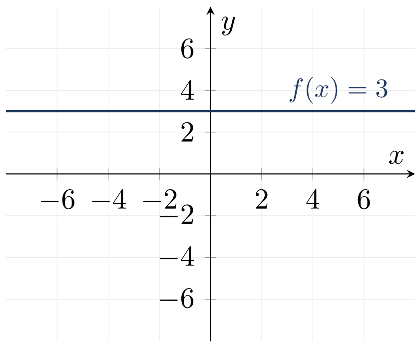
Identity Function

Let $\mathbb{R}$ be the set of real numbers. Define the real valued function $f : \mathbb{R} \to \mathbb{R}$ by $y = f(x) = x$ for each $x \in \mathbb{R}$. Such a function is called the *identity function*. Here the domain and range of f are $\mathbb{R}$.



The graph is a straight line passing through the origin.

Define the function $f : \mathbb{R} \to \mathbb{R}$ by $y = f(x) = c$, $x \in \mathbb{R}$, where c is a constant for each $x \in \mathbb{R}$. Such a function is called the *constant function*. Domain $= \mathbb{R}$, Range $= \{c\}$.



The graph is a line parallel to the x-axis.

Polynomial Function

A function $f : \mathbb{R} \to \mathbb{R}$ is said to be a polynomial function if for each x in $\mathbb{R}$,

$$y = f(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n,$$

where n is a non-negative integer and $a_0, a_1, a_2, \ldots, a_n \in \mathbb{R}$.

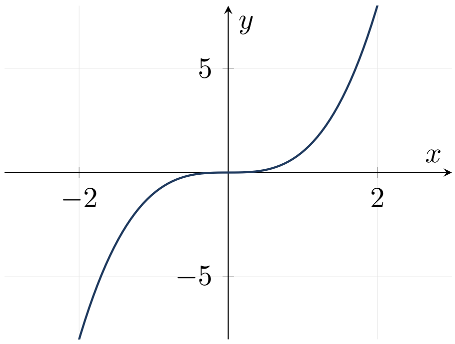
Examples:

$$f(x) = x^3 - x^2 + 2, \qquad g(x) = x^4 + \sqrt{2}\,x$$

Classwork 32. $h(x) = x^{2/3} + 2x$ is not a Polynomial function. Why?

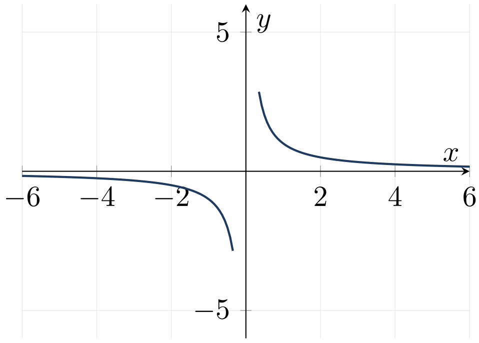
Rational Functions

The graph of the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3$, $x \in \mathbb{R}$. Domain $= \mathbb{R}$, Range $= \mathbb{R}$.



Rational functions are functions of the type $\dfrac{f(x)}{g(x)}$, where $f(x)$ and $g(x)$ are polynomial functions of x defined in a domain, where $g(x) \neq 0$.

The real valued function $f : \mathbb{R} - \{0\} \to \mathbb{R}$ defined by $f(x) = \dfrac{1}{x}$, $x \in \mathbb{R} - \{0\}$. Domain $= \mathbb{R} - \{0\}$, Range $= \mathbb{R} - \{0\}$.



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Relations and Functions

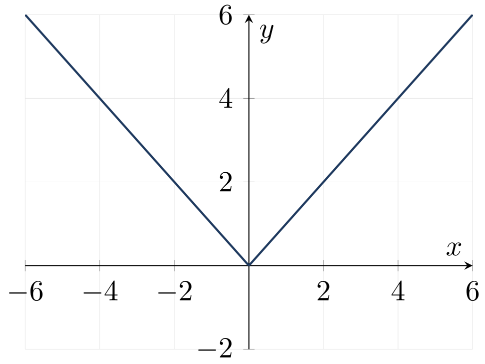
DAY 06

The Modulus Function

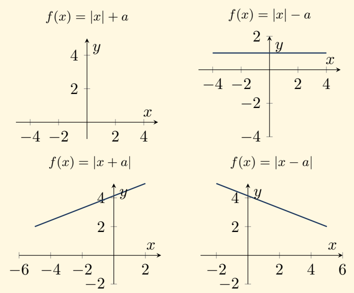
The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = |x|$ for each $x \in \mathbb{R}$ is called the *modulus function*.

$$f(x) = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

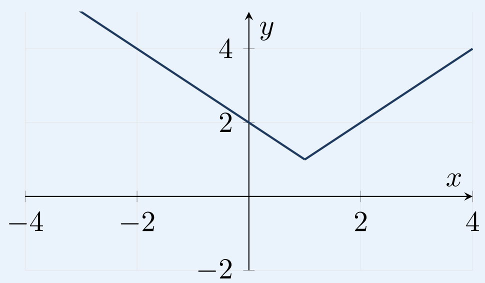
Domain $= \mathbb{R}$, Range $= [0, \infty)$.



Note



Classwork 33. Graph of the function $f : \mathbb{R} \to \mathbb{R}$ is given below:



(i) Write the value of $f(2)$.
(ii) Identify the function and choose the correct answer.
(A) $f(x) = |x + 1|$ (B) $f(x) = |x - 1|$
(C) $f(x) = |x| + 1$ (D) $f(x) = |x| - 1$

Classwork 34. Consider the function $f(x) = |x| - 3$, draw the graph of $f(x)$. Write the domain and range of $f(x)$.

Classwork 35. Draw the graph of $f(x) = -|x|$. [NCERT]

Classwork 36. Draw the graph of $f(x) = 3 - |x|$.

Classwork 37. Sketch the graph of the function $f(x) = |x + 1|$.

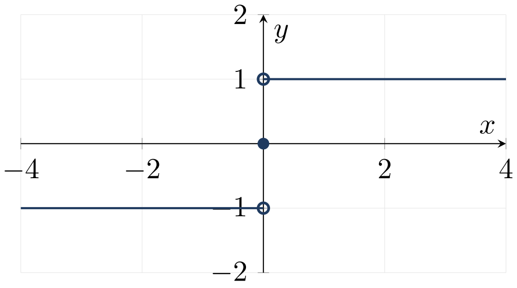
Signum Function

The function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -1, & \text{if } x < 0 \end{cases}$$

Domain $= \mathbb{R}$, Range $= \{-1, 0, 1\}$.
Or

$$\operatorname{sgn}(x) = \begin{cases} \dfrac{|x|}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$



Classwork 38. Draw the graph of the function

$$f(x) = \begin{cases} 2, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -3, & \text{if } x < 0 \end{cases}$$

Write domain and range of the above function.

Classwork 39. If f is a signum function, then consider a real valued function $f(100) = \ldots$

Classwork 40. The range of the function $f(x) = \dfrac{|x-4|}{x-4}$ is

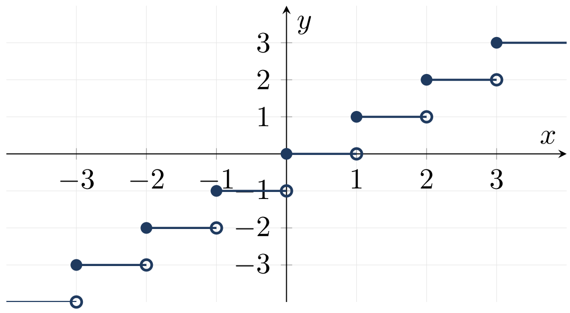
(A) ϕ (B) $\mathbb{R}$ (C) $\{1\}$ (D) $\{-1\}$ (E) $\{1, -1\}$

Greatest Integer Function

The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = [x]$, $x \in \mathbb{R}$, assumes the value of the greatest integer, less than or equal to x. Such a function is called the *greatest integer function*.

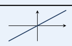
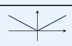
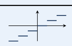
Example

$[x] = -1$ for $-1 \leq x < 0$
$[x] = 0$ for $0 \leq x < 1$
$[x] = 1$ for $1 \leq x < 2$
$[x] = 2$ for $2 \leq x < 3$



Domain $= \mathbb{R}$, Range $= \mathbb{Z}$.

Classwork 41. Match the following:

	Graph		Function
i)		1)	Modulus function: $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = \lvert x \rvert$
ii)		2)	Signum function: $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = \begin{cases} \frac{\lvert x \rvert}{x}; & x \neq 0 \\ 0; & x = 0 \end{cases}$
iii)		3)	Identity function: $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = x$
iv)		4)	Greatest integer function: $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = [x]$

Classwork 42. Draw the graph of the function

$$f(x) = \begin{cases} 1 + x, & -1 \leq x \leq 0 \\ 1 - x, & 0 < x \leq 1 \end{cases}$$

Classwork 43. The function f is defined by

$$f(x) = \begin{cases} 1 - x, & x < 0 \\ 1, & x = 0 \\ x + 1, & x > 0 \end{cases}$$

Draw the graph of $f(x)$. [NCERT]

Classwork 44. Draw the graph of the function $f(x) = (x - 1)^2$.

Homework 9. The function f defined by

$$f(x) = \begin{cases} 1 + x & \text{if } x < 0 \\ 1 & \text{if } x = 0 \\ 1 - x & \text{if } x > 0 \end{cases}$$

Draw the graph of $f(x)$.

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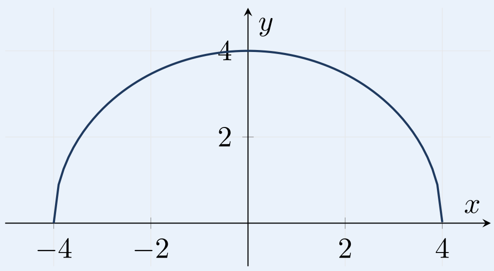
Relations and Functions

DAY 07

Domain and Range From Graph of a Function

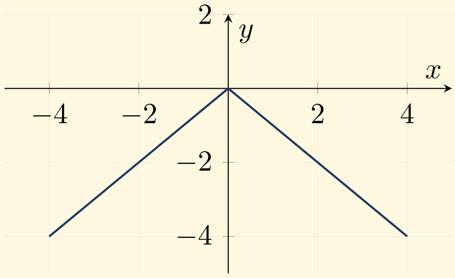
Classwork 45. The figure shows the graph of a function $f(x)$ which is a semi-circle centred at origin.



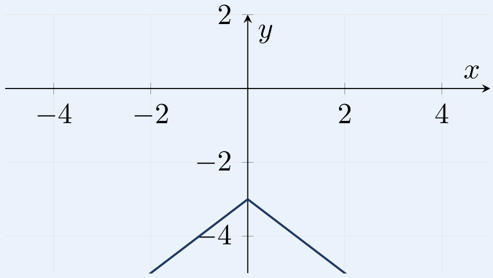
Write the domain and range of $f(x)$.

Classwork 46. Find the domain and range of the function $f(x) = -|x|$. [NCERT]

Note

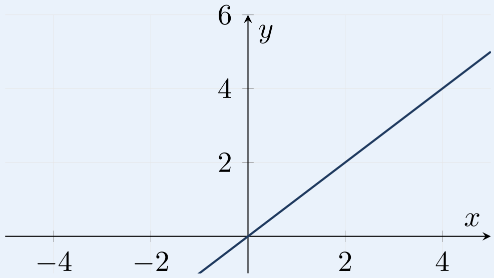


Classwork 47. Find domain and range of the function given below.



Algebra of Real Functions

Classwork 48. The figure shows the graph of the function $f(x)$.



Write the domain and range of $f(x)$.

Homework 10. Consider the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = |x - 2|$.
(i) The value of $f(0) =$
(A) 0 (B) 2 (C) -2 (D) 1
(ii) Draw the graph of the function f.
(iii) Hence write the range of the function f.

Addition of two real functions:

Let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be any two real functions, where $X \subset \mathbb{R}$. Then, we define $(f + g) : X \to \mathbb{R}$ by

$$(f + g)(x) = f(x) + g(x) \quad \text{for all } x \in X.$$

Subtraction of a real function from another:

Let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be any two real functions, where $X \subset \mathbb{R}$. Then, we define $(f - g) : X \to \mathbb{R}$ by

$$(f - g)(x) = f(x) - g(x), \quad \text{for all } x \in X.$$

Multiplication of two real functions:

The product (or multiplication) of two real functions $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ is a function $fg : X \to \mathbb{R}$ defined by

$$(fg)(x) = f(x)\,g(x), \quad \text{for all } x \in X.$$

This is also called *pointwise multiplication*.

Quotient of two real functions:

Let f and g be two real functions defined from $X \to \mathbb{R}$, where $X \subset \mathbb{R}$. The quotient of f by g, denoted by $\frac{f}{g}$, is a function defined by

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, \quad \text{provided } g(x) \neq 0,\ x \in X.$$

Domain and Range of Functions in the Form $f(x) = \dfrac{1}{x - a}$

Classwork 49. Let $f(x) = x^2$ and $g(x) = 2x + 1$ be two real functions. Find $(f + g)(x)$, $(f - g)(x)$, $(fg)(x)$, $\left(\frac{f}{g}\right)(x)$. [NCERT]

Classwork 50. Find the domain of the function $f(x) = \dfrac{1}{x - 9}$.

Homework 11. Let $f(x) = \sqrt{x}$ and $g(x) = x$ be two functions defined over the set of non-negative real numbers. Find $(f+g)(x)$, $(f-g)(x)$, $(fg)(x)$ and $\left(\frac{f}{g}\right)(x)$. [NCERT]

Homework 12. The domain of the function $f(x) = \dfrac{1}{x-1}$ is

(A) $\{1\}$ (B) $\mathbb{R}$ (C) $\mathbb{R} - \{1\}$ (D) $\mathbb{R} - \{0\}$

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Relations and Functions

DAY 08

Domain and Range of Functions in the Form $f(x) = \sqrt{f(x)}$, where $f(x) \geq 0$

Classwork 51. Consider the function $f(x) = \sqrt{x-1}$, find domain and range of f. [NCERT]

Classwork 52. Find the domain and range of $f(x) = \sqrt{9 - x^2}$. [NCERT]

Domain and Range of Functions in the Form $f(x) = \dfrac{p(x)}{q(x)},\ q(x) \neq 0$

Classwork 53. Let $g(x) = 2 - 3x$, $x \in \mathbb{R}$, $x > 0$ and $h(x) = x^2 - 3x + 2$, $x \in \mathbb{R}$ be two functions. Find:
(i) Range of $g(x)$
(ii) Domain of $\dfrac{g(x)}{h(x)}$

Classwork 54. Consider a real valued function $f(x) = \dfrac{x-3}{x^2 - x - 6}$. Find the domain of $f(x)$.

Classwork 55. Find the domain and range of function $f(x) = \dfrac{x^2 + 3x + 5}{x^2 - 5x + 4}$.

Homework 13. Consider the real function $f(x) = \dfrac{x+2}{x-2}$.
(i) Find the domain and range of the function.
(ii) Prove that $f(x) \cdot f(-x) + f(0) = 0$.

Case Study

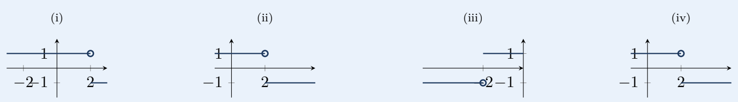
I. Read the following passage and answer the questions given below.
A teacher asked randomly his two students to draw graph (figure) on the black board. Two students Raman and Prashant draw figure (graphs) (a) and (b) one by one on the board as shown below:

(i) For the figure (a), write the function in terms of x.
(ii) (a) For the figure (b), write the function in terms of x.
 (b) Write the range of the function $f(x)$ of the figure (a).
OR
(iii) The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = |x|$ then write the value of $f(-3)$.

2. The function 't' which maps temperature in degree Celsius into temperature in degree Fahrenheit is defined by $t(C) = \dfrac{9C}{5} + 32$. Find
(i) $t(0)$ (ii) $t(28)$ (iii) $t(-10)$ (iv) The value of C, when $t(C) = 212$ [NCERT]

3. Find the domain and range of $f(x) = \sqrt{25 - x^2}$. [NCERT]

4. Consider the function $f(x) = \dfrac{x-2}{|x-2|}$
(a) Identify the graph of the function f.



(b) Write the domain and range of f.

Try Yourself

1. If $P = \{m, n\}$, $Q = \{n, m\}$; State whether the following is TRUE or FALSE.
$P \times Q = \{(m, n), (n, m)\}$

Try Yourself

5. (a) If $A = \{a, b\}$, write $A \times A \times A$.
(b) If $R = \{(x, x^3) : x$ is a prime number, less than $10\}$. Write R in roster form.
(c) Find the domain and range of the function $f(x) = 2 + \sqrt{x - 1}$.

6.

Fig. 1

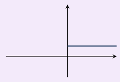



Fig. 2

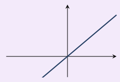



Fig. 3

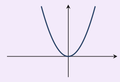
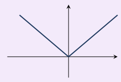



Fig. 4


(i) Which of the above graph doesn't represent a function?
(ii) Identify signum function from above graphs.
(iii) Write domain and range of signum function.

7. Find the range of the function $f(x) = [2x]$, $-\dfrac{1}{3} \leq x \leq \dfrac{8}{3}$.

8. Consider the real valued function $f(x) = \dfrac{x+2}{x-2}$
(a) Find the domain and range of the function.
(b) Prove that $f(x) \cdot f(-x) + f(0) = 0$.

9. Find the domain and the range of the real function f defined by $f(x) = \sqrt{(x - 1)}$. [NCERT]

10. Find the range of the function $f(x) = x^2 + 2$, x is a real number. [NCERT]

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